

BRUNEI INTL, Brunei Darussalam

WMO: 963150

Lat: 4.944N

Lon: 114.928E

Elev: 73

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.2	73.8	69.3	107.9	86.1	70.1	111.0	84.8	18.5	83.3	16.7	83.0	4.8	190	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.1	92.9	78.4	91.7	78.7	91.1	78.8	81.8	88.0	81.3	87.5	80.8	87.1	8.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.3	85.0	79.2	152.2	84.0	79.1	151.5	83.9	45.7	88.4	45.1	88.3	44.6	87.5	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	13.5	12.0	70.2	95.9	3.8	2.1	67.5	97.4	65.3	98.7	63.2	99.9	60.5	101.4
			69.3	83.9	3.5	1.8	66.7	85.2	64.7	86.3	62.7	87.3	60.2	88.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	81.4	81.6	82.5	83.2	83.6	83.1	82.8	82.9	82.7	82.1	82.0	81.8
	DBStd	1.81	1.77	1.72	1.56	1.48	1.68	1.76	1.76	1.90	1.73	1.69	1.40	1.67
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11848	973	884	1008	996	1041	992	1017	1019	980	994	959	985
	CDD65	6373	508	464	543	546	576	542	552	554	530	529	509	520
	CDH74	67483	5034	4637	5777	6161	6563	5945	5919	5998	5660	5383	5160	5246
	CDH80	24761	1520	1351	2010	2443	2691	2357	2329	2393	2171	1926	1806	1764
(14) Wind	WSAvg	5.3	5.9	6.3	5.6	5.0	4.9	5.0	5.0	5.3	5.2	5.2	5.2	5.3
(15) Precipitation	PrecAvg	115.0	10.6	5.8	5.1	7.3	9.2	8.1	8.7	9.1	10.3	12.5	13.3	14.3
	PrecMax	145.4	24.7	17.4	13.2	17.6	20.6	24.7	17.9	28.0	28.4	22.9	20.3	28.2
	PrecMin	88.5	0.1	0.3	0.0	2.0	2.7	2.7	1.9	0.7	2.5	1.1	5.7	3.7
	PrecStd	15.7	6.9	4.8	3.3	3.9	4.5	4.3	4.1	5.9	5.5	4.4	4.0	6.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	90.7	92.0	93.0	93.4	93.2	93.4	93.5	92.9	91.7	91.4	91.2
		MCWB	78.2	77.6	78.2	79.4	79.5	77.7	77.9	77.8	78.3	78.5	78.5	78.7
	2%	DB	88.2	88.3	89.9	91.2	91.7	91.6	91.7	91.7	91.3	90.0	89.7	89.6
		MCWB	78.5	78.5	78.7	79.6	79.7	78.5	78.4	78.2	78.5	78.7	79.1	78.7
	5%	DB	87.6	87.5	88.8	89.8	91.0	90.3	90.1	90.6	89.8	89.3	88.5	88.2
		MCWB	78.5	78.6	78.8	79.5	79.7	78.7	78.6	78.3	78.6	78.8	79.1	79.0
	10%	DB	86.2	86.1	87.7	89.1	89.6	89.4	89.2	89.4	88.8	87.9	87.7	87.4
		MCWB	78.2	78.2	78.8	79.5	79.4	78.8	78.7	78.5	78.7	78.8	79.0	78.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	80.8	81.3	81.9	82.6	81.8	81.8	81.9	81.8	81.4	81.3	81.2
		MCDB	85.7	86.1	87.5	88.1	89.1	87.7	88.5	87.8	88.0	88.2	87.0	86.6
	2%	WB	80.0	79.9	80.4	81.4	81.7	80.9	80.8	80.9	80.7	80.4	80.4	80.4
		MCDB	84.9	85.1	86.2	87.6	88.2	87.2	87.4	87.2	87.3	86.8	86.5	85.9
	5%	WB	79.3	79.2	79.9	80.7	81.0	80.2	80.0	80.1	80.0	79.8	79.9	79.8
		MCDB	84.2	84.2	85.4	87.0	87.5	86.6	86.5	86.4	86.5	86.0	85.8	85.2
	10%	WB	79.0	78.9	79.2	80.1	80.4	79.7	79.4	79.4	79.4	79.2	79.3	79.2
		MCDB	83.8	83.7	84.6	86.1	86.8	86.0	85.8	85.8	85.7	85.2	85.0	84.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	10.4	11.8	12.7	13.1	13.2	13.4	13.5	13.2	12.6	12.4	11.7
		MCDBR	12.1	11.6	13.0	13.5	14.1	14.3	14.7	14.9	14.4	14.0	13.4	12.9
		MCWBR	4.3	4.1	4.5	4.7	4.9	4.9	5.2	5.2	5.3	4.9	4.9	4.6
	5% WB	MCDBR	11.1	10.8	12.0	12.8	13.1	13.3	13.5	13.6	13.3	13.0	12.7	12.1
		MCWBR	4.3	4.2	4.5	4.9	5.1	5.1	5.4	5.3	5.5	5.1	5.0	4.7
(40) Clear-Sky Solar Irradiance	taub		0.414	0.424	0.427	0.426	0.408	0.411	0.409	0.447	0.448	0.457	0.414	0.412
	taud		2.502	2.449	2.441	2.442	2.506	2.505	2.505	2.374	2.369	2.352	2.493	2.511
	Ebn at Noon		288	287	285	280	279	274	276	271	275	274	285	286
	Edh at Noon		35	37	38	37	34	33	34	39	40	41	35	34
(44) All-Sky Solar Radiation	RadAvg		1512	1705	1814	1828	1704	1612	1620	1660	1621	1531	1538	1490
	RadStd		139	125	100	74	72	71	76	65	97	102	92	109

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	N/A	N/A	N/A	+0.31	+0.32	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.27	N/A	N/A	+0.52	+0.47	+0.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page